Denoising-Time Steering Along a Paired Skin-Tone Direction in SDXL: A Preregistered Matched-Change Study

Anonymous submission

Abstract

We study whether a linear direction estimated from paired portrait latents can control visible skin tone in Stable Diffusion XL while preserving other image properties. The proposed method computes a paired mean difference in the VAE latent space, applies a spatial Gaussian mask, and distributes the resulting perturbation across denoising steps. We preregister a matched-change evaluation after calibration, estimate the direction from 64 of 96 noise-coupled prompt pairs, validate an image-colour instrument on the remaining 32 pairs, and evaluate 570 conditions over 30 held-out seeds. All images generated; 569 of 570 conditions had complete metrics. At a matched five-degree Individual Typology Angle (ITA) change, masked steering showed no detectable advantage in face-embedding similarity: every confidence interval included zero. Compared with unmasked steering, however, masking reduced LPIPS by 0.023 in the lighter direction and 0.082 in the darker direction (Holm-adjusted $p = .012$ and $p = .002$). In the darker direction it also improved background SSIM relative to unmasked steering and improved LPIPS, background SSIM, and pose relative to prompt-only generation. Matched-target coverage was 63%/87% for masked, 43%/50% for unmasked, and 23%/60% for prompt-only steering in the lighter/darker directions. The result supports a localized perceptual-preservation claim, not identity preservation, disentanglement, or bias mitigation. Skin tone is treated only as an apparent visual attribute, never as a proxy for race or ethnicity.

1 Introduction

Latent diffusion models synthesize high-resolution images by denoising in a compressed representation rather than directly in pixel space [7]. SDXL extends this family with a larger U-Net and richer conditioning [6]. These models offer expressive control but also make it difficult to isolate one visual attribute without changing identity, pose, or composition.

Linear semantic directions have been effective in GAN latent spaces [9]. We ask whether an analogous direction, estimated from paired SDXL VAE encodings, can be used during diffusion sampling. Our technical hypothesis is that a denoiser can absorb small perturbations injected throughout sampling more coherently than a single post-hoc edit. Our scientific hypothesis is narrower: at a matched change in measured skin tone, stepwise masked injection will preserve non-target properties better than relevant baselines.

We make three contributions. First, we specify the paired estimator and denoising-time intervention. Second, we provide a preregistered matched-change comparison that separates target response from preservation. Third, we release an auditable research artifact with immutable model and data hashes, measurement validation, seed-level uncertainty, missingness, and failures.

2 Related work

This study does not claim that semantic directions in diffusion models are new. Asyrp identifies a semantic representation space inside frozen diffusion U-Nets and manipulates directions across the reverse process [3]; later work discovers concept-specific internal directions for text-to-image generation [4]. Concept Sliders instead learn low-rank parameter directions for continuous, composable control [1]. Our narrower object of study is a direction formed directly from paired SDXL VAE encodings, added to scheduler latents throughout sampling without optimization. The empirical question is whether that simple intervention offers a preservation advantage at matched target response, not whether diffusion representations admit semantic control in general.

3 Construct and scope

Race and ethnicity are social identities, not one-dimensional visual attributes. The intervention in this study is therefore called a *skin-tone direction*. It must not be interpreted as changing, estimating, or classifying race. Even datasets designed to balance face categories expose the difficulty and consequences of reducing social categories to visual labels [2]. The present method is intended for controlled generative-model research and auditing, not profiling or decisions about people.

4 Method

Let $E(x)$ be the deterministic posterior-mode encoding of an image under the SDXL VAE. Given n noise-coupled prompt pairs (x_i^L, x_i^D) generated with the same seed and prompt template but different skin-tone descriptors, the raw direction is

$$v = \frac{1}{n}\sum_{i=1}^{n}\left[E(x_i^D) - E(x_i^L)\right]. \tag{1}$$

Seed reuse controls the initial diffusion noise but does not guarantee the same apparent identity after conditioning changes. Identity, composition, prompt wording, and illumination therefore remain potential training-direction confounders.

We multiply v by a spatial mask $M \in [0,1]^{H\times W}$ centered on the face:

$$M(p) = w_e + (w_c - w_e)\exp\left[-3\left(\frac{\|p\|_2}{r}\right)^2\right], \quad \tilde{v} = M \odot v, \tag{2}$$

where the study uses center weight $w_c = 1$, edge weight $w_e = 0.3$, and radius $r = 1$. For T reverse-diffusion steps and steering magnitude α, the pipeline updates the scheduler latent after each step by

$$z_t \leftarrow z_t + \frac{\alpha}{T}\tilde{v}. \tag{3}$$

The implementation uses 25 DPM-Solver++ multistep updates, motivated by fast diffusion ODE solvers [5], with classifier-free guidance 7.5.

5 Evaluation design

The target response and preservation outcomes must be measured separately. The continuous target proxy uses cheek-pixel CIE Lab individual typology angle (ITA). Instrumental ITA decreases as measured pigmentation increases [12], but apparent skin color in images is affected by hue, illumination, and acquisition conditions and is not fully represented by a single light–dark axis [10]. We therefore validate ordering and prespecified illumination perturbations on a held-out split and describe the result only as apparent image color.

The primary preservation outcome is face-embedding similarity, following the general embedding approach of FaceNet [8], at a matched continuous skin-tone change. Secondary outcomes are LPIPS perceptual distance [13], background-only SSIM [11], head-pose drift, face-detection failure, and monotonicity.

The confirmatory design used held-out seeds and four methods: prompt-only generation, post-hoc latent addition, unmasked stepwise injection, and masked stepwise injection. The preregistered configuration contains 30 held-out generation seeds and method-specific grids frozen after calibration. Comparisons are paired within seed. We report means, standard deviations, medians, detector failures, and 95% seed-level bootstrap confidence intervals; secondary confirmatory tests use Holm correction. A run cannot pass when a required metric is missing.

Table 1: Held-out validation of the continuous image-colour instrument.

Criterion	Required	Observed
Detection rate	≥ 0.95	1.000
Pair-order accuracy	≥ 0.90	0.969
Median pair gap (ITA)	≥ 8.0	16.67
Median perturbation shift	≤ 2.0	0.40
95th-percentile shift	≤ 5.0	3.06

6 Pre-confirmatory validation and calibration

The direction-data campaign generated 96 noise-coupled light/dark prompt pairs. We reserve 64 pairs for direction estimation and 32 pairs for measurement validation. The ITA instrument detected the face and cheek regions in all 32 held-out pairs, ordered the light-descriptor image above the dark-descriptor image in 31 of 32 pairs (96.9%), and produced a median within-pair separation of 16.67 ITA degrees. Across prespecified brightness and white-balance perturbations, the median absolute ITA shift was 0.40 degrees and the 95th percentile was 3.06 degrees. Table 1 shows that every frozen gate passed before confirmatory generation.

Calibration used seeds disjoint from both the direction-data and confirmatory sets. The frozen target was an absolute five-degree ITA change with a three-degree matching tolerance. Prompt-only, unmasked stepwise, and masked stepwise steering achieved both directions for all three calibration seeds. Post-hoc addition achieved the darker target for all three seeds but the lighter target for only one; wider endpoint probes remained non-monotonic and identity-dependent. We therefore froze post-hoc addition as a descriptive feasibility and dose-response baseline, excluding it from the primary matched contrast before confirmatory generation.

7 Confirmatory results

7.1 Completion and target response

The runner produced all 570 prespecified condition rows with 570 unique keys and the frozen configuration fingerprint. All images generated. One post-hoc condition (seed 500019, $\alpha = -1.5$) lacked a detectable edited face/skin region, leaving 569 metric-complete rows (99.8%). The failure remains in the missingness denominator. All 330 conditions in the primary matched-change methods were complete.

Table 2: Prespecified matched-target coverage over 30 seeds.

Method	Lighter	Darker
Prompt-only	7 (23%)	18 (60%)
Stepwise, unmasked	13 (43%)	15 (50%)
Stepwise, masked	19 (63%)	26 (87%)

Figure 1 shows the mean target response. Mean seed-level Spearman correlations between α and skin-tone change were 0.95 for prompt-only, 0.93 for unmasked stepwise, 0.92 for masked stepwise, and 0.67 for post-hoc addition. The post-hoc median was 0.99 but its mean 95% interval was $[0.43, 0.88]$, confirming substantial seed dependence.

The target was reached within the three-degree tolerance unevenly (Table 2). Masked steering had the highest coverage in both directions. Among successful matches, its mean achieved changes were 5.06 ITA lighter and 5.23 ITA darker. Low prompt-only lighter coverage constrains that stratum to six seeds shared with the masked reference.

7.2 Preservation at matched change

The primary face-similarity hypothesis was not supported. Masked-stepwise advantages relative to prompt-only were -0.0008 (95% CI $[-0.0035, 0.0016]$, $n = 6$) for lighter edits and 0.0032 ($[-0.0056, 0.0149]$, $n = 17$) for darker edits. Relative to unmasked stepwise steering, the advantages were 0.0020 ($[-0.0014, 0.0056]$, $n = 13$) and 0.0079 ($[-0.0027, 0.0184]$, $n = 14$). All Holm-adjusted p values were 1.0.

Masking nevertheless improved perceptual preservation relative to unmasked steering: the paired LPIPS advantage was 0.0235 ($[0.0128, 0.0360]$, $p_{\mathrm{Holm}} = .012$) for lighter edits and 0.0825 ($[0.0476, 0.1302]$, $p_{\mathrm{Holm}} = .002$) for darker edits. Against prompt-only generation, the darker-direction LPIPS advantage was 0.0812 ($[0.0379, 0.1218]$, $p_{\mathrm{Holm}} = .028$). The corresponding darker-direction improvements were 0.0207 in background SSIM ($[0.0105, 0.0319]$, $p_{\mathrm{Holm}} < .001$) and 1.61 in the pose-difference proxy ($[0.88, 2.36]$, $p_{\mathrm{Holm}} = .007$). Masking also improved background SSIM by 0.0129 versus unmasked darker edits ($[0.0080, 0.0176]$, $p_{\mathrm{Holm}} = .002$). No other secondary contrast survived Holm correction. Figure 2 shows every prespecified matched preservation contrast, including null and adverse estimates.

Figure 3 is illustrative rather than inferential. Its seed was selected by a fixed rule: maximize the number of complete matched rows, then choose the seed closest to the median base ITA, breaking ties by lower seed. The prompt-only example visibly demonstrates why matched image-colour change does not imply matched identity or composition.

Table 3: Exploratory split-half direction stability. Entries are median cosine similarity with 2.5th–97.5th percentiles over 200 disjoint resamples.

Pairs/half	Raw direction	Masked direction
8	0.42 [0.28, 0.53]	0.53 [0.38, 0.63]
16	0.59 [0.50, 0.66]	0.69 [0.60, 0.75]
32	0.75 [0.69, 0.78]	0.82 [0.77, 0.84]

8 Post-confirmatory robustness

We performed two explicitly exploratory checks after completing the frozen analysis. First, we repeated matching with one- and two-degree ITA tolerances. The LPIPS advantage retained the same positive direction in every estimable stratum, and face-similarity estimates remained compatible with no advantage. However, shared matched samples collapsed to $n = 1$–5 at one degree and $n = 3$–12 at two degrees; no secondary contrast survived Holm correction. Thus, the preregistered three-degree results should not be interpreted as evidence of precisely matched effects under stricter tolerances.

Second, we re-encoded all 64 training pairs and confirmed that the resulting direction was exactly equal to the recorded confirmatory tensor (maximum absolute difference zero). We then drew 200 random disjoint split-halves at each pair count. Table 3 reports cosine agreement between independently estimated directions. Stability increased with sample size, and masking improved agreement, but even two disjoint 32-pair estimates had median cosine 0.75 before masking and 0.82 after masking. The direction is therefore reproducible under the fixed 64-pair campaign but not sample-invariant.

9 Limitations

The study evaluates one generator, scheduler, resolution, and model revision; cross-model generalization is unknown. Synthetic direction pairs can encode prompt wording, brightness, age, gender presentation, hair, or facial features alongside skin tone, and seed reuse does not preserve identity. A center Gaussian is not a semantic skin mask. The same generator creates direction and evaluation images, so model-specific artifacts may dominate. ITA is an image-colour proxy, not a complete representation of skin appearance, and its validation uses synthetic prompt pairs rather than calibrated physical colour measurements. Face embeddings and face/skin detectors can exhibit demographic error. Matched-change coverage is incomplete and particularly low for prompt-only lighter edits, reducing paired sample sizes and precision. The post-confirmatory tolerance analysis confirms that stricter matching is too sparse for stable inference, while the split-half analysis shows that the learned direction remains sensitive to which synthetic pairs are sampled.

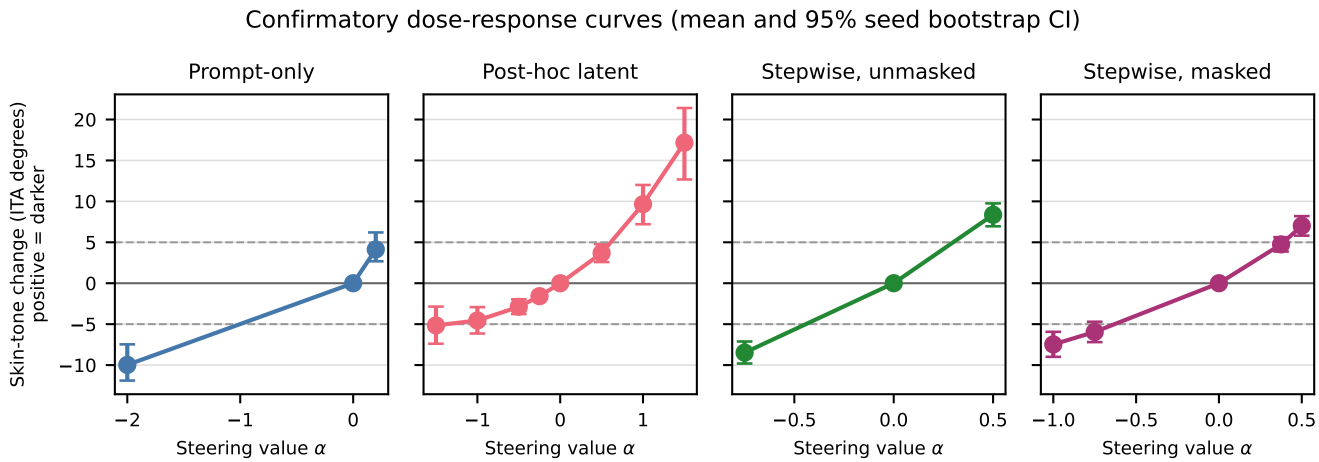

Figure 1: Confirmatory dose-response curves. Points are seed means and bars are 95% seed-level bootstrap intervals. Dashed lines mark the prespecified ±5 ITA targets; positive change denotes a darker measured appearance.

The pose measure is a five-landmark proxy rather than a full 3D estimator. No human study assesses perceived identity, realism, or social harms. Finally, the additive update is scheduler-dependent and has not been validated across model or library revisions.

10 Conclusion

Across 30 held-out seeds, masked denoising-time injection offered more reliable target coverage than the tested unmasked and prompt-only grids and improved LPIPS preservation relative to unmasked steering in both directions. Several darker-direction background and pose contrasts also favored masking. The preregistered primary face-similarity hypothesis was not supported, so the result must not be described as identity preservation, disentanglement, or bias mitigation. The principal contribution is a bounded empirical result and an auditable protocol for separating target response, preservation, missingness, and unsupported social claims.

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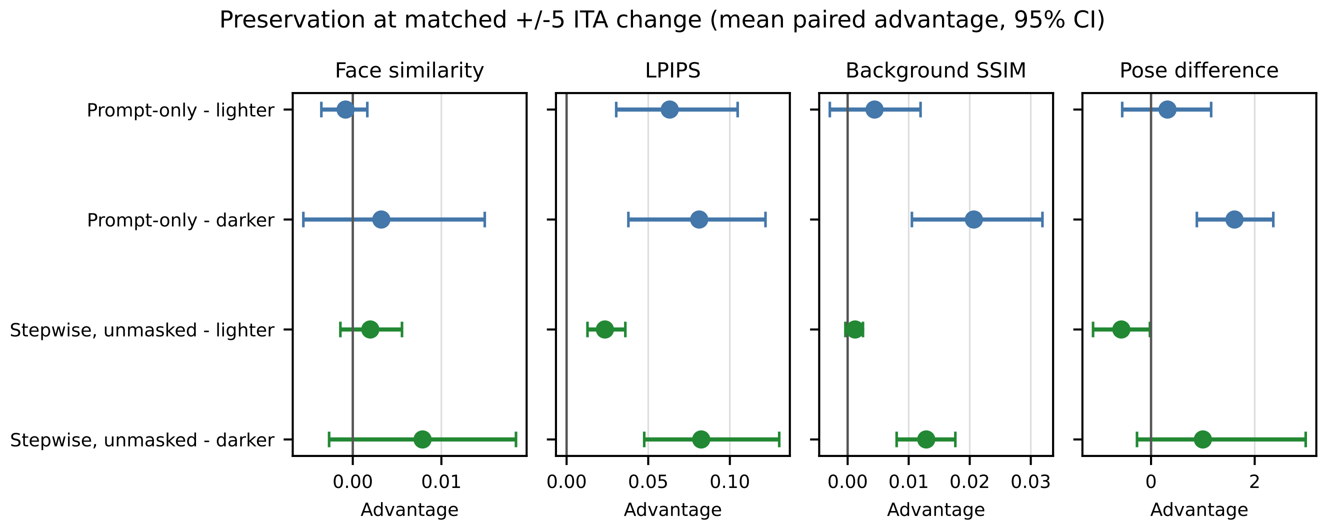

Figure 2: Masked-stepwise preservation advantage at matched ±5 ITA change. Bars are 95% seed-level bootstrap intervals. Positive values favor masked steering after orienting every outcome so that larger is better.

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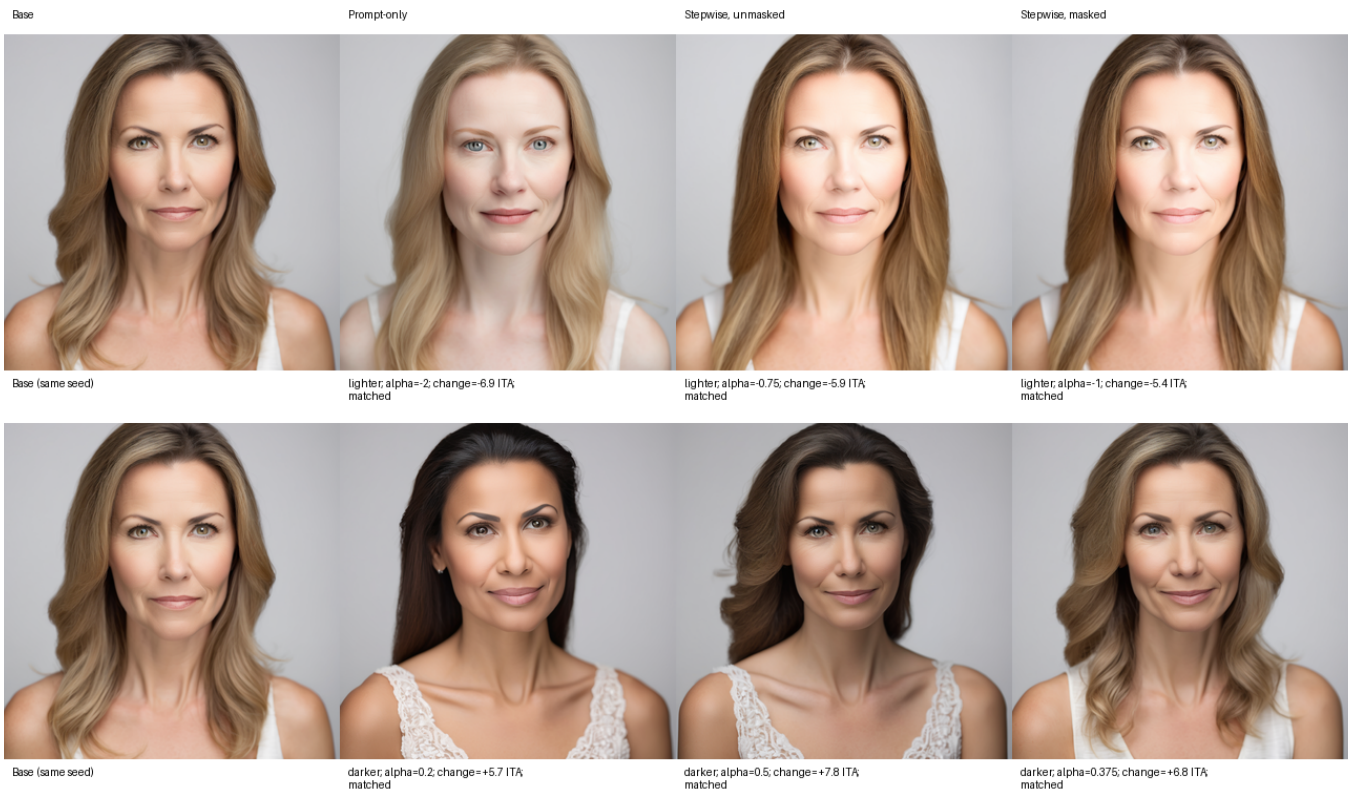

Figure 3: Deterministically selected confirmatory example (seed 500011). Each row reuses the same base image; labels report the selected alpha and achieved ITA change. Prompt-only conditioning can alter apparent identity despite reusing the diffusion seed.